Ajay Devidas Wanekar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.79
Vasantrao Naik Jr. Collage of Science and Arts, Cidco, Ch. Sambhajinagar, Maharashtra	Class XII (Maharashtra State Board)	2017	67.38%
Maharashtra Public School, DTP, Ch. Sambhajinagar	Class X (Maharashtra State Board)	2014	83.00%

WORK EXPERIENCE

Project Associate | LFOVIA Lab, IIT Hyderabad

Evaluated and fine-tuned Vision-Language Models (VLMs), including Florence-2 Base/Large, Llama 3.2 Vision, and Qwen2-VL, for multimodal text-image understanding, and benchmarked task-specific models including RetinaFace, DeepFace, and Siamese architectures for targeted detection, face analysis, and visual tracking/re-identification.

Junior Research Assistant | TIHAN, IIT Hyderabad:

Worked on multimodal maritime perception by developing and curating near-infrared and sonar datasets, implementing sensor-specific preprocessing, annotation, data-quality optimization, and model-training pipelines for robust object detection in complex maritime environments, resulting in an IEEE conference publication.

Research Intern | WINET Lab, IIT Hyderabad:

Developed foundational sonar-based underwater object-detection datasets by implementing annotation, Python-based preprocessing, data augmentation, and markup pipelines to improve data quality and model robustness, with the work published at the IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2023).

PROJETS

Multi-Channel Conversational AI Platform:

- Built a multi-tenant conversational AI platform unifying voice, chat, and email with real-time STT/TTS pipelines and multilingual support for low-latency customer-care interactions.

- Developed a hybrid RAG pipeline (semantic + keyword retrieval) with LLM-based intent classification, improving answer relevance by 18% for customer data interactions.

Multimodal Vision-Language Model Evaluation for Computer Vision Tasks:

- Evaluated Vision-Language Models (VLMs) for multimodal (text-image) understanding across computer vision tasks, including object detection, segmentation, age and gender classification, crowd analysis, and action/activity recognition, as alternatives to conventional task-specific detectors.

- Curated and annotated RGB and depth-mapping datasets, building structured preprocessing, labeling, and quality-validation pipelines that standardized inputs and improved model training consistency and reliability.

- Benchmarked and fine-tuned models using Python-based pipelines, evaluating detection performance (precision/recall, IoU) and optimizing data-processing workflows for improved scalability and computational efficiency.

M.TECH THESIS

Hybrid Machine Learning Modeling for Multi-Objective Optimization of a Non-Oxidative Methane Coupling DBD Reactor

- Developed and benchmarked Linear Regression, Random Forest, XGBoost, and Gaussian Process Regression models to predict DBD reactor performance from operating conditions, reducing dependence on costly experimental trials.

- Engineered features from a 24-sample experimental dataset using one-hot encoding and evaluated model generalization through Leave-One-Out Cross-Validation (LOOCV).

- Achieved cross-validated R^2 values up to 0.996 using Gaussian Process Regression across six reactor-performance outputs, including CH conversion, H selectivity, and C-C hydrocarbon selectivities.

- Performed feature-importance analysis and assessed model limitations related to extrapolation and confounded variables, identifying Bayesian Optimization as a potential strategy for experiment-guided process optimization.

TECHNICAL SKILLS

Programming & Frameworks: Python, C++, SQL, PyTorch, CUDA, Pandas, NumPy, Scikit-learn, TensorFlow

AI & ML: Computer Vision (CV), Vision-Language Models, Large Language Models (LLMs), LLM Fine-Tuning, ANN, CNN, YOLO, DeepFace, RetinaFace, Multimodal Data Processing (Text, Image) Transformer Architectures

CONFERENCE PUBLICATIONS

[1] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. "Novel Approach to Underwater Object Detection Using Sonar Sensors for Autonomous Underwater Vehicles (AUVs)." IEEE i-COSTE 2023, pp. 1-5.

[2] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. "Object Detection and Classification for Autonomous Surface Vehicles (ASVs) Through Near-Infrared Imaging." IEEE i-COSTE 2023, pp. 1-6.

HACKATHONS & ACHIEVEMENTS

- Top 10 All India - Winner, Meta x Pragati AI Hackathon.

- AI and X Hackathon, IIT Hyderabad participant